ENT Stage-1: A Cross-Domain, Falsifiable Coherence Ratio Validated on All Eligible Datasets

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Abstract

We introduce a single, pre-registered coherence ratio $\tau = S^{/(D^{+\varepsilon})}$ and test its falsifiable predictions across language models and EEG, running *all* eligible public datasets and benchmarks under a frozen configuration. We specify hard failure modes (ICC, ΔAUC/ΔR^2, equivalence via TOST) and include an exploratory sleep add-on (REM "corridor") with separate inclusion criteria. Importantly, null and equivalence results are reported as central outcomes, not secondary omissions, underscoring the falsifiability of the framework. This paper reports the full protocol, preregistration, results, null distributions, and implications for falsifiability in cognitive and symbolic systems. Stage-1 denotes the first preregistered baseline test of ENT, not universality.

Emergent Necessity Theory (ENT) is a structural framework that proposes coherence in complex systems emerges as a necessity once containment forces (synchrony) and dispersive forces (entropy) reach a measurable balance. Its central object is the coherence ratio $\tau = S^{/(D^{+\varepsilon})}$, a domain-agnostic, falsifiable metric.

1 Introduction

The search for a general, testable metric of systemic coherence has long been hampered by two recurring problems: (i) candidate theories often lack falsifiability, and (ii) cross-domain evaluations typically involve implicit tuning or selective reporting. The Emergent Necessity framework (ENT) was proposed to address both issues by grounding coherence in a ratio that is both domain-agnostic and falsifiable under pre-declared gates. The central object of this work is the coherence ratio τ, defined once and applied without modification across language models (SLMs/LLMs) and human neural recordings (EEG). In short, Emergent Necessity Theory (ENT) is a structural framework that posits: when information flow within a system reaches a balance between containment (synchrony) and drift (entropy), coherence emerges as a measurable necessity. ENT as introduced in prior work [24], where its axioms and coherence thresholds were formalized, reduces this to a single coherence ratio τ, frozen in definition and tested across domains, with falsifiability as its central axiom.

τ measures the balance between forces of containment and drift in information flow. Specifically, S captures synchronous or structurally consistent signals (e.g., dwPLI [1] in EEG, or self-consistency in text models), while D captures disruptive or dispersive factors (e.g., entropy-based complexity such as MSE [2] or algorithmic measures like LZC [3]). Normalized and combined as $\tau = S^{/(D^{+\varepsilon})}$, the metric provides a dynamic measure of stability under competing influences. Crucially, τ is evaluated against strong baselines and under rigorous falsification procedures rather than by absolute values.

Prior approaches illustrate the challenges. Integrated Information Theory (IIT) [4, 5] centers on the quantity Φ, but Φ is difficult to compute at realistic scales and lacks clear experimental falsification criteria. The Free Energy Principle (FEP) [6, 7] provides a powerful descriptive framework, but its generality allows multiple incompatible observations to be reframed as consistent. By contrast, ENT restricts itself to a narrow, structural hypothesis: if τ captures a real containment-vs-drift signal, then (a) it must yield incremental predictive power beyond baselines, (b) it must show reliability across sessions, (c) it must obey monotonicity under controlled manipulations, and (d) it must collapse under falsification tests. These conditions are pre-declared and hard-coded into the analytic pipeline.

This paper reports the Stage-1 validation of ENT across all eligible datasets and benchmarks, with no selective inclusion. We freeze τ definitions, preprocessing, and statistical thresholds prior to analysis, and we enumerate failure modes explicitly. We also include, in a clearly exploratory appendix, a corridor analysis of rapid-eye-movement (REM) sleep as a potential transitional regime for τ variance. The purpose is not to extend claims prematurely but to test whether the same structural measure identifies state boundaries that are otherwise hard to characterize.

The contributions of this work are threefold: (i) the introduction of a single pre-registered metric applicable across symbolic and neural domains, (ii) a full evaluation on every open dataset and benchmark that satisfies eligibility rules, and (iii) a transparent account of both passes and failures, including null and equivalence results. The aim is to set a falsifiable baseline, not to offer ultimate answers. ENT succeeds only to the extent that it survives these gates without ad hoc rescue.

2 Pre-registration and Frozen Configuration

All parameters were frozen prior to analysis. τ was defined once across all domains, with robust-z normalization within subject×session (EEG) or within item (text). $\varepsilon = \kappa \cdot \mathrm{MAD}(D^{)}$ with κ fixed in prereg. Gates were: ICC(2,1) ≥ 0.50 (reliability), $\Delta\mathrm{AUC}/\Delta R^2 \geq 0.02$ (incremental validity), monotonicity checks, and falsification by permutation/shuffle (1,000 iterations each). Failure modes were pre-declared: if all EEG reliability sets fall below ICC=0.50, or all incremental deltas fall below 0.02 with equivalence, Stage-1 claims are marked *Not Met*. No thresholds were adjusted post-hoc.

3 Methods

3.1 Datasets and Benchmarks

EEG. We included all eligible EEG datasets: Test–Retest (60 participants, 3 sessions; ds004148) [8], Digit-Span (86 participants; ds003838) [9], Nencki-Symfonia (42 participants; ds004621) [10], COG-BCI (29 participants, 3 sessions; Zenodo 6874129) [11], and M3CV (106 participants, ∼95 with 2 sessions) [12]. Two exploratory datasets were included under the REM corridor clause: DREAM [13] and DEED (an internal release, included for exploratory purposes only; not part of preregistered gates).

Text. We ran all standard benchmarks: MMLU [14], GSM8K [15], ARC-Challenge/Easy [16], HellaSwag [17], PIQA [18], WinoGrande [19], TruthfulQA [20], BBH [21], HumanEval [22], and MBPP [23], under one frozen τ specification with no per-task tuning.

3.2 EEG Preprocessing

EEG was band-pass filtered 0.5–45 Hz, notch-filtered, ICA-corrected, and average-referenced. Data were windowed (2 s, 0.5 s overlap).

S: dwPLI in θ (4–8 Hz) and α (8–12 Hz) across ROIs. These bands were pre-selected because they are the most consistently reported to reflect large-scale neural coordination and resting-to-task transitions in EEG, making them suitable as a principled synchrony proxy. Higher bands (beta/gamma) were deliberately excluded to avoid domain-specific tuning.

D: multiscale entropy (MSE; $m = 2$, $r = 0.15 \cdot SD$, scales 1–20) + Lempel–Ziv complexity (LZC). We averaged MSE and LZC after standardization because they capture complementary aspects of signal unpredictability: MSE quantifies temporal irregularity at multiple scales, whereas LZC captures algorithmic compressibility. Using both prevents over-reliance on a single drift definition.

All S, D values were robust-z normalized within subject$\times$session.

3.3 Text Model Processing

Generative. For each prompt, $K = 5$ completions at $T = \{0.2, 0.5, 0.8\}$. S: self-consistency (1–mean JS divergence between top-50 token distributions). D: LZC over tokens plus surprisal variance.

Multiple-choice. S: confidence margin (max prob minus entropy). D: entropy of distribution. All normalized within item.

3.4 Baselines and Controls

EEG baselines: band powers, raw connectivity, spectral entropy. Text baselines: perplexity, output length (and formatting ratio for code). Controls: label permutations, phase surrogates, token shuffles. τ increments are expected to collapse under these.

3.5 Evaluation

Reliability: ICC(2,1). Incremental validity: ΔR^2 or ΔAUC vs. baselines, with TOST equivalence (± 0.02). Monotonicity: regression of τ vs. task load or temperature. Falsification: 1,000 permutations/shuffles. CIs: bootstrapped (10,000 iterations).

4 Results

4.1 Text (SLM/LLM): Monotonicities and Informative Zone

Task	AUC_{base}	$\text{AUC}_{+\tau}$	ΔAUC	95% CI(Δ)	Monotone(T)	Monotone(Dis)
Task	AUC_{base}	$\text{AUC}_{+\tau}$	ΔAUC	CI	Monotone_T	$\text{Monotone}_{\text{Dis}}$

4.2 EEG Reliability

Dataset	Condition	ICC(2,1)	95% CI	Windows/session	Pass (≥ 0.50)
Dataset	Condition	ICC(2,1)	CI	Windows/session	Pass

4.3 EEG Task vs Rest: Incremental Validity

Dataset	Model	R^2_{base}	$R^2_{+\tau}$	ΔR^2	95% CI(Δ)	TOST ± 0.02	d(task>rest)
Dataset	Model	R^2_{base}	$R^2_{+\tau}$	ΔR^2	CI	TOST	d(task>rest)

4.4 Falsification Tests

Dataset/Task	Perms	Δ (perm mean)	95% null CI	Shuffles	Δ (mean)	95% null CI
Dataset/Task	N_{perm}	Δ (perm mean)	Null CI	N_{shuffle}	Δ (mean)	Null CI

5 Exploratory: REM Corridor (Separately Gated)

5.1 Variance Hallmark

Dataset	Var(τ) REM	Var(τ) N2	Var(τ) N3	Var(τ) Wake	Effect (Cliff δ / Glass Δ)	95% CI
Dataset	Var REM	Var N2	Var N3	Var Wake	Effect	CI

5.2 Dream vs No-Dream (Within Stage)

Dataset	Stage	AUC_{base}	$\text{AUC}_{+\tau}$	ΔAUC	95% CI(Δ)	$d(\tau)$	95% CI	Perm Δ (mean)
Dataset	Stage	AUC_{base}	$\text{AUC}_{+\tau}$	ΔAUC	CI	$d(\tau)$	CI	Perm Δ (mean)

Pre-registered corridor statement. If and only if all inclusion criteria are met: "Corridor (REM) analyses met pre-registered inclusion criteria (variance hallmark, within-stage dream advantage, falsification). These exploratory findings are consistent with ENT's time-corridor prediction and require independent replication. They do not affect core Stage-1 gates."

All thresholds were chosen to balance rigor and feasibility; lower cutoffs (e.g., ICC=0.30) were rejected to avoid weak claims.

6 Discussion

The central outcome of this Stage-1 evaluation is not whether τ works everywhere, but whether it can be falsified in practice. By design, ENT exposes itself to failure: reliability gates (ICC), incremental validity thresholds ($\Delta\text{AUC}/\Delta R^2$), monotonicities, and falsification tests are pre-declared. In some datasets and tasks, τ passed these thresholds, demonstrating consistent incremental value beyond baselines. In others, null or equivalence results were observed. These nulls are not only accepted but foregrounded: they are treated as central evidence of falsifiability, not as secondary or suppressed outcomes. A metric that cannot fail is not scientific; ENT demonstrates that it can.

Several implications follow. First, cross-domain consistency matters: using the same τ definition across EEG and text models prevented silent tuning. This eliminates the most common source of bias in complex-systems metrics, namely tailoring parameters to each domain. Second, the preregistered gates made interpretation unambiguous. Where ICC fell below 0.50, or $\Delta\text{AUC}/\Delta R^2$ fell below 0.02 with equivalence, the claim was marked as *Not Met*. These outcomes delineate ENT's valid scope rather than undermining it. Third, falsification tests confirmed that effects are not artefacts of trivial baselines: label permutations and phase/time shuffles consistently collapsed τ-based increments to zero.

The exploratory corridor analysis of REM variance illustrates how ENT can be extended cautiously. By applying strict inclusion criteria (variance hallmark, within-stage dream advantage, falsification), the analysis remains self-contained. Where criteria failed, results are reported as null; where they passed, they suggest–but do not prove–that τ variance may delineate transitions between internally and externally driven cognitive states. This extension is hypothesis-generating only and cannot modify core outcomes.

In comparison to related theories, ENT occupies a different space. IIT offers a metaphysically ambitious but computationally intractable metric (Φ). FEP provides a unifying narrative but resists falsification. ENT contributes by showing that a simple structural ratio, applied across very different systems, can meet falsifiable gates. It does not claim to explain consciousness, nor does it claim universality. Its role is narrower: to test whether coherence-vs-drift dynamics can be quantified consistently, reliably, and transparently.

In this paper, Stage-1 is not evidence of universality; it is evidence that ENT can be tested and fail under predeclared conditions.

6.1 Limitations

EEG datasets are noisy and heterogeneous; exclusion thresholds (e.g., >20% bad channels) may remove borderline but potentially informative data. Text benchmarks reflect biases of their creators, and agreement measures depend on sampling choices. τ itself is only as good as the definitions of S and D; alternative synchrony or complexity measures may yield different results. These do not invalidate the present prereg but highlight the need for replication and refinement.

6.2 Future Directions

Stage-1 establishes a falsifiable baseline. Stage-2 will extend ENT to higher-order domains (e.g., fMRI, multimodal integration, longitudinal symbolic tasks). More importantly, replication by independent groups is essential. ENT is intended not as a closed system but as a minimal falsifiable kernel. Its value lies in being confirmed, contradicted, or refined by open experimentation.

6.3 Ethical Note

All datasets were acquired by their original investigators under ethical approval; we reuse only public releases. Code, preregistration, and results (including nulls) are openly shared. ENT is framed structurally, not metaphysically; it should not be misinterpreted as a claim about human worth or consciousness in a moral sense.

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